

Distributional Extension Taxonomy

Regular Dominated

Examples: Gaussian location, Gaussian scale
Local W2: regular metric scaling
Testing: quadratic KL / Hellinger
Role: theorem target

Support-Changing Nonregular

Examples: uniform scale
Local W2: can remain linear
Testing: nonquadratic / one-sided KL
Role: separate theory needed

Singular / Atomic

Examples: two-point scale
Local W2: can remain linear
Testing: singular
Role: exclusion / counterexample class

Transport-Hölder control alone is too weak for a universal extension theorem. The correct theorem classes are regular dominated families and operational geometries with analogous local structure.